

The Bell Curve

[Code ▾](#)

Foundational Concepts

Well, the most famous shape of distributions, ever, is the bell-shaped curve, which is called the Gaussian or the normal distribution. So, the family of curves—the normal curves or the Gaussian curves—is the topic of this lecture.

[Hide](#)

```
if (!caracas::has_sympy()) {
  caracas::install_sympy()
}
library(caracas)
library(sets)
library(tibble)
library(ggplot2)
library(dplyr)
```

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```
X = c(1:10)
Σ = function(xs, n=0) sum(xs, n)
μ = function(xs) Σ(xs)/length(xs)
σ_2 = function (xs) Σ((xs - μ(xs))^2/ length(xs)
P = function(E, p) E/p
f = function(x=x) x
C = function(n, k) factorial(n)/(factorial(k)*factorial((n - k)))
def_sym("x", "y", "z", "n", "p", "k", "Lambda", "e", "sigma", "Sigma", "mu")
```

Two Gaussian distributions can differ in their means (centers). For example, in the histograms of the heights of men and the heights of women, the means of the two just shifted over. Two Gaussian distributions can differ in their spread, as well. For example, the batting averages in 1920 and 2000 are both approximately normal curves with approximately the same mean but differing in their spread or standard deviation.

The normal distribution has a specific mathematical formula. The normal distribution's formula has a formidable and complicated look:

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```
normal_sym = 1/(sigma*sqrt(2*pi)) * e^(-1/2*((x - mu)/sigma)^2)
normal_sym
```

$$c: \frac{0.398942280401433 \cdot e^{-0.5 \cdot (-\mu + x)^2 / \sigma^2}}{\sigma}$$

One interesting feature is that it contains both of the most famous constants in mathematics, and e. It has two parameters, traditionally labeled by the Greek letters (mu) and (sigma). Any particular choices for and ($\mu > 0$) determine a specific distribution; that is, we would know its shape exactly. Setting $\mu = 0$ and $\sigma = 1$ gives the standard normal curve. The number is the mean of the distribution. The number is the standard deviation, which measures the spread.

Approximations of the normal distribution arise when we start with any shaped distribution and look at the distribution of averages of samples of a certain size. We will illustrate this idea with an example using a standard deck of cards. Assign a value to each card: Ace = 1, 2 = 2, 3 = 3, ..., 10 = 10, Jack = 11, Queen = 12, and King = 13. For each possible five-card hand in the whole deck, take the average value of the cards in the hand.

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```
numbers_vector = 1:9
faces_vector = c("Ace", "Jack", "Queen", "King")

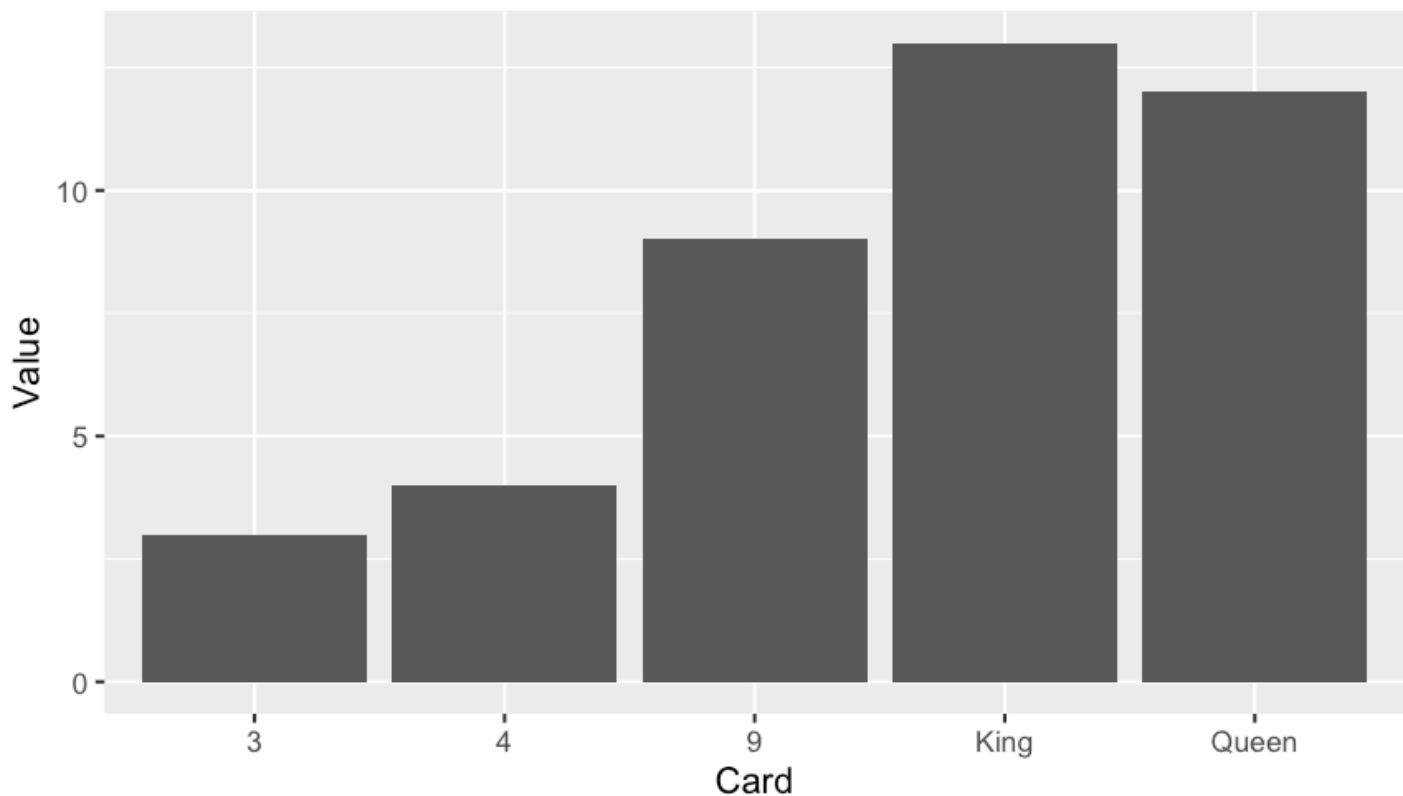
card_deck_df <- tibble(card = c(faces_vector, numbers_vector), value = c(10, 11, 12, 13, numbers_vector))
```

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```
card_hand_df <- card_deck_df |>
  slice_sample(n = 5)

ggplot(card_hand_df, aes(x = card, y = value)) +
  geom_col() +
  labs(
    title = "My Card Hand",
    x = "Card",
    y = "Value"
  )
```

My Card Hand



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```
deck_mean <- mean(card_deck_df$value)
deck_mean
```

```
[1] 7
```

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```
sample_means_df_generator <-
  function(card_deck_df = card_deck_df, hand_size = 5, num_simulations = 10000) {
    tibble(means = purrr::map_dbl(1:num_simulations, \(x) card_deck_df |>
      slice_sample(n = hand_size, replace = FALSE) |>
      pull(value) |> mean()))
  }
sample_means_df_generator(card_deck_df)
```

means
<dbl>

7.8

7.6

8.6

6.2

9.0

6.4

5.4

8.2

8.4

7.6

1-10 of 10,000 rows

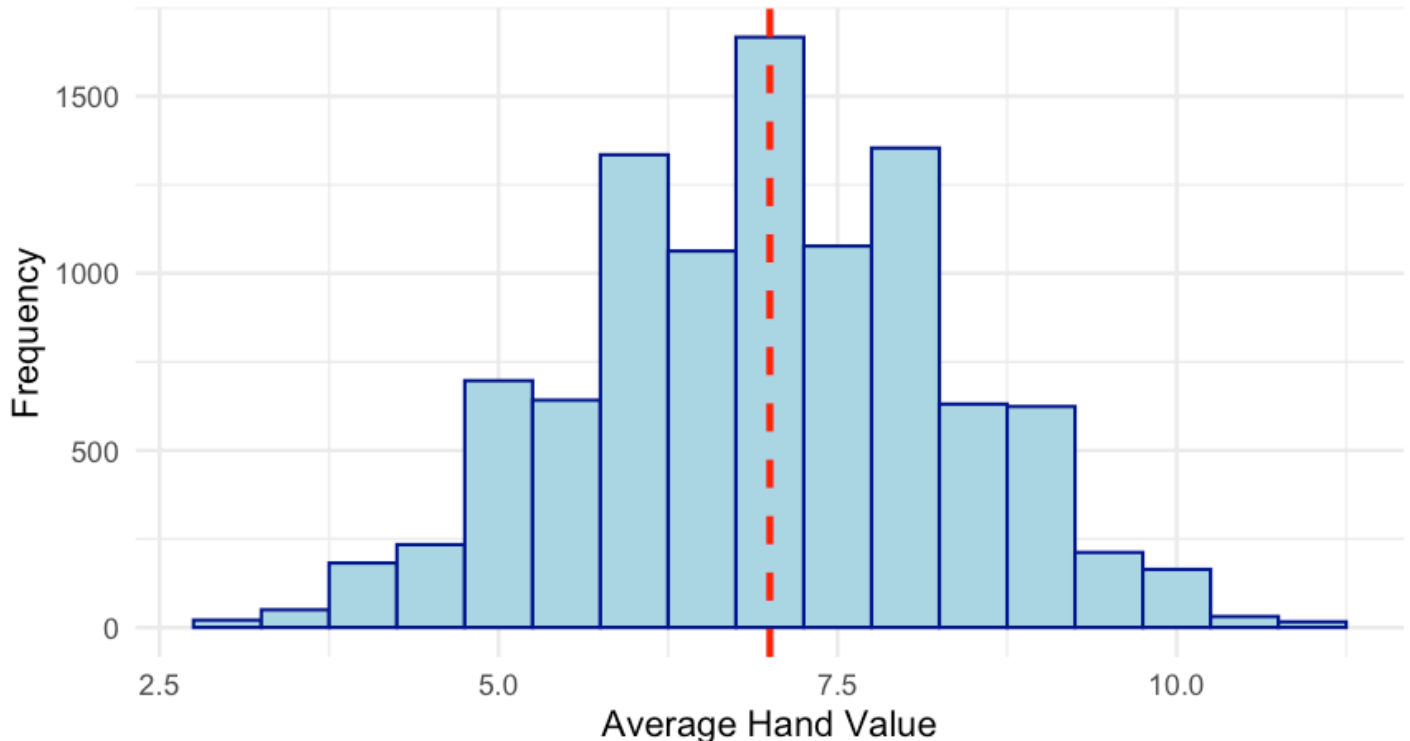
Previous 1 2 3 4 5 6 ... 100 Next

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```
ggplot(sample_means_df_generator(card_deck_df), aes(x = means)) +  
  geom_histogram(binwidth = 0.5, fill = "lightblue", color = "darkblue") +  
  geom_vline(xintercept = deck_mean, linetype = "dashed", color = "red", linewidth  
= 1) +  
  labs(  
    title = "Distribution of Sample Means (Central Limit Theorem)",  
    subtitle = paste("Based on", 10000, "hands of size", 5),  
    x = "Average Hand Value",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```

Distribution of Sample Means (Central Limit Theorem)

Based on 10000 hands of size 5



The central limit theorem says that starting with (almost) any distribution (such as a Poisson distribution, or a binomial distribution, or a uniform distribution), if we take many samples of size n , the distribution of the average values of the samples will be approximately a Gaussian distribution (assuming n is large). The

standard deviation of the sample population will be the original population's standard deviation divided by the square root of n . This fact will be useful in doing statistical inference. The approximation gets better and better as the sample size n gets larger and larger.

Concept and Applications

COVARIANCE AND CORRELATION

if you're new to statistics, you may be ready to jump on the cause-and-effect bandwagon when you find a strong relationship between 2 variables. But have you ever thought about why 2 variables might be correlated? So far, when we've considered variance, we've limited ourselves to 1 variable. But what if we have 2 variables that we think might be related? How might they vary together? This brings us to the idea of covariance, and from there to correlation.

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```
X = c(2,3,5,6,8,9,10,13)
Y = c(58,75,71,77,80,88,83,95)
```

COVARIANCE

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```
Cov <- function(X, Y, n = length(X)) sum((X - mean(X)) * (Y - mean(Y)) / (n - 1))

Cov(X,Y)
```

```
[1] 38
```

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```
cov(X, Y)
```

```
[1] 38
```

CORRELATION

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```
Cor(X, Y)
```

```
[1] 0.9173286
```

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```
 $\sigma_{2\_sample}(X)$ 
```

```
[1] 10.27778
```

CORRELATION AND COVARIANCE

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```
# not done yet!  
data("Harman23.cor")  
head(Harman23.cor$cov)
```

	height	arm.span	forearm	lower.leg	weight	bitro.diameter	chest.girth
chest.width 0.382	1.000	0.846	0.805	0.859	0.473	0.398	0.301
arm.span 0.415	0.846	1.000	0.881	0.826	0.376	0.326	0.277
forearm 0.345	0.805	0.881	1.000	0.801	0.380	0.319	0.237
lower.leg 0.365	0.859	0.826	0.801	1.000	0.436	0.329	0.327
weight 0.629	0.473	0.376	0.380	0.436	1.000	0.762	0.730
bitro.diameter 0.577	0.398	0.326	0.319	0.329	0.762	1.000	0.583